# 11933 - JWST spectroscopy of the next bright kilonova: mapping heavy element nucleosynthesis

Cycle: 5, Proposal Category: GO

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**OBSERVATIONS**

| *Folder* | *Observation* | *Label* | *Observing Template* | *Science Target* |
|---|---|---|---|---|
| Observation Folder | | | | |
| | 1 | NIRSpec 17d | NIRSpec Fixed Slit Spectroscopy | (1) Kilonova |
| | 2 | NIRCam 17d | NIRCam Imaging | (1) Kilonova |
| | 3 | MIRI LRS 17d | MIRI Low Resolution Spectroscopy | (1) Kilonova |
| | 4 | MIRI Imaging 17d | MIRI Imaging | (1) Kilonova |
| | 5 | NIRSpec 30d | NIRSpec Fixed Slit Spectroscopy | (1) Kilonova |
| | 6 | NIRCam 30d | NIRCam Imaging | (1) Kilonova |
| | 7 | MIRI LRS 30d | MIRI Low Resolution Spectroscopy | (1) Kilonova |
| | 8 | MIRI Imaging 30d | MIRI Imaging | (1) Kilonova |
| | 9 | NIRSpec 60d | NIRSpec Fixed Slit Spectroscopy | (1) Kilonova |
| | 10 | NIRCam 60d | NIRCam Imaging | (1) Kilonova |
| | 11 | MIRI Imaging 60d | MIRI Imaging | (1) Kilonova |

**ABSTRACT**

Many elements heavier than iron are thought to be synthesised by the rapid neutron-capture process (r-process), but which environments have the appropriate temperatures, energies and free neutron fluxes has long been a mystery. In recent years, it has become clear that neutron star mergers produce radioactive r-process elements that power thermal kilonovae, which subsequently decay into stable isotopes of the heavy elements that we see around us today. The question of whether such kilonovae are the dominant source of trans-iron nuclei remains open. Near-infrared spectra of two kilonovae (one associated with a gravitational wave detection and both with a gamma-ray burst) contain spectral features that resemble emission lines from optically thin ejecta, but further examples are urgently required to map their diversity, and better establish feature identifications. New developments in atomic data calculations for heavy elements and non-LTE radiative transfer models allow us to calculate realistic synthetic spectra for kilonovae in the nebular regime. Predictions of strong, forbidden line transitions provide testable model predictions which can quantify the composition of the merger ejecta, and constrain hydrodynamic simulations of neutron star mergers. Only JWST has the sensitivity and spectral wavelength coverage in the infrared to test these simulations. We propose a non-disruptive ToO for three epochs of spectroscopy of a kilonova to compare with sophisticated spectral synthesis models. We will measure the elements formed in neutron star mergers, thus quantitatively constraining their role in the chemical evolution of the r-process elements in the Universe.

**OBSERVING DESCRIPTION**

We propose three epochs of contemporaneous NIRSpec, NIRCam and MIRI observations of the next nearby kilonova, at approximately 17, 30 and 60 days post-merger. Such kilonovae are likely to be identified via some combination of gravitational waves, gamma-rays, X-rays, or high-cadence optical sky surveys. Our observing strategy has been carefully tuned from state-of-the-art radiative transfer simulations to maximise scientific return from each epoch of observation.

Our first epoch (~17 days) commences with a 1 hour observation with NIRSpec prism. This observation will be accompanied by 10 minute exposures in each of six wide NIRCam filters (F115W, F150W, F200W, F277W, F356W and F444W), which will allow crucial calibration of the NIRSpec continuum. We will also perform observations with MIRI, both imaging and low-resolution spectroscopy (LRS). We will observe with LRS for 3 hours, immediately followed by 15 minute exposures in each of four wide MIRI filters (F560W, F770W, F1000W and F1280W), again to facilitate critical spectral continuum calibration.

Our second epoch (~30 days) follows an identical observing strategy, but with extended exposure times to account for transient evolution. Our NIRSpec exposure time will increase to 2 hours, and each NIRCam exposure will increase to 35 minutes. We will observe with MIRI LRS for 4.7 hours, followed by 30 minute exposures for each MIRI wide filter.

Our third epoch of observation (~60 days) follows a similar observing strategy, but without MIRI LRS, as the kilonova is too faint to return useable data. Our NIRSpec exposure increases to 6 hours, and each NIRCam exposure will increase to 55 minutes. We persist with MIRI imaging; for each MIRI wide filter, we will obtain 90 minute exposures.

These data will enable us to constrain the evolution of the kilonova ejecta properties in detail.

**Generic Targets**

| # | Name | Criteria | Description |
|---|---|---|---|
| (1) | Kilonova | The kilonova transient we will target | |

## Observation

**Proposal 11933, Observation 1: NIRSpec 17d**

Fri Mar 13 22:05:24 GMT 2026

**Diagnostic Status: Warning**

Observing Template: NIRSpec Fixed Slit Spectroscopy

## Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Generic Targets

| # | Name | Criteria | Description |
|---|---|---|---|
| (1) | Kilonova | The kilonova transient we will target | |

## Acquisition

| # | Target | TA Method | Subarray | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | SAME | WATA | FULL | CLEAR | NRSRAPIDD6 | 3 | 1 | 1 | 171.788 | 274128 |

## Template

| HFF Readout Mode | Slit | Subarray |
|---|---|---|
| false | S400A1 | FULL |

## Dithers

| # | Primary Dither Positions | Sub-Pixel Pattern |
|---|---|---|
| 1 | 2 | SPATIAL |

## Spectral Elements

| # | Grating/Filter | Slit | Readout Pattern | Groups/Int | Integrations/Exp | # | Autocal | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | PRISM/CLEAR | S400A1 | NRSIRS2RAPID | 60 | 1 | 1 | NONE | 4 | 4 | 3559.689 | 274128 |

| Special Requirements | |
|---|---|
| | Target Of Opportunity Response Time 14 Days |

## Observation

**Proposal 11933, Observation 2: NIRCam 17d**

Fri Mar 13 22:05:25 GMT 2026

**Diagnostic Status: Warning**

Observing Template: NIRCam Imaging

## Diagnostics

(NIRCam 17d (Obs 2)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Generic Targets

| # | Name | Criteria | Description |
|---|---|---|---|
| (1) | Kilonova | The kilonova transient we will target | |

## Template

| Module | Subarray | Target Placement |
|---|---|---|
| ALL | FULL | Module gap (large extended source) |

## Dithers

| # | Primary Dither Type | Primary Dithers | Subpixel Dither Type | Dither Size | Subpixel Positions |
|---|---|---|---|---|---|
| 1 | INTRAMODULEX | 4 | STANDARD | | 1 |

## Spectral Elements

| # | Short Filter | Long Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Dithers | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|
| 1 | F115W | F277W | SHALLOW4 | 3 | 1 | 4 | 4 | 601.259 | 274128 |
| 2 | F150W | F356W | SHALLOW4 | 3 | 1 | 4 | 4 | 601.259 | 274128 |
| 3 | F200W | F444W | SHALLOW4 | 3 | 1 | 4 | 4 | 601.259 | 274128 |

## Special Requirements

Target Of Opportunity Response Time 14 Days

## Observation

**Proposal 11933, Observation 3: MIRI LRS 17d**

Fri Mar 13 22:05:25 GMT 2026

**Diagnostic Status: Warning**

Observing Template: MIRI Low Resolution Spectroscopy

## Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Generic Targets

| # | Name | Criteria | Description |
|---|---|---|---|
| (1) | Kilonova | The kilonova transient we will target | |

## Acquisition

| # | Target | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 Kilonova | F560W | FASTGRPAVG16 | 10 | 1 | 1 | 444.006 | 274128 |

## Template

| Subarray | Obtain Verification Image? |
|---|---|
| FULL | false |

## Dithers

| # | Dither Type | No. Spectral Steps | Spectral Step Offset | No. Spatial Steps | Spatial Step Offset |
|---|---|---|---|---|---|
| 1 | ALONG SLIT NOD | | | | |

## Spectral Elements

| # | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Exposures/Dith | Total Dithers | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|
| 1 | FASTR1 | 100 | 20 | 40 | 1 | 2 | 11205.612 | 274128 |

| Special Requirements | |
|---|---|
| | Target Of Opportunity Response Time 14 Days |

## Observation

**Proposal 11933, Observation 4: MIRI Imaging 17d**

Fri Mar 13 22:05:25 GMT 2026

**Diagnostic Status: Warning**

Observing Template: MIRI Imaging

## Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Generic Targets

| # | Name | Criteria | Description |
|---|---|---|---|
| (1) | Kilonova | The kilonova transient we will target | |

## Template

| Subarray |
|---|
| FULL |

## Dithers

| # | Dither Type | Starting Point | Number of Points | Points | Starting Set | Number of Sets | Optimized For | Direction | Pattern Size |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2-Point | | | | | | | | DEFAULT |

## Spectral Elements

| # | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Exposures/Dith | Dither | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | F560W | FASTR1 | 80 | 2 | 1 | Dither 1 | 2 | 4 | 893.563 | 274128 |
| 2 | F770W | FASTR1 | 80 | 2 | 1 | Dither 1 | 2 | 4 | 893.563 | 274128 |
| 3 | F1000W | FASTR1 | 80 | 2 | 1 | Dither 1 | 2 | 4 | 893.563 | 274128 |
| 4 | F1280W | FASTR1 | 80 | 2 | 1 | Dither 1 | 2 | 4 | 893.563 | 274128 |

## Special Requirements

Target Of Opportunity Response Time 14 Days

## Observation

**Proposal 11933, Observation 5: NIRSpec 30d**

Fri Mar 13 22:05:25 GMT 2026

**Diagnostic Status: Warning**

Observing Template: NIRSpec Fixed Slit Spectroscopy

## Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Generic Targets

| # | Name | Criteria | Description |
|---|---|---|---|
| (1) | Kilonova | The kilonova transient we will target | |

## Acquisition

| # | Target | TA Method | Subarray | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | SAME | WATA | FULL | CLEAR | NRSRAPIDD6 | 3 | 1 | 1 | 171.788 | 274128 |

## Template

| HFF Readout Mode | Slit | Subarray |
|---|---|---|
| false | S400A1 | FULL |

## Dithers

| # | Primary Dither Positions | Sub-Pixel Pattern |
|---|---|---|
| 1 | 2 | SPATIAL |

## Spectral Elements

| # | Grating/Filter | Slit | Readout Pattern | Groups/Int | Integrations/Exp | # | Autocal | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | PRISM/CLEAR | S400A1 | NRSIRS2RAPID | 120 | 1 | 1 | NONE | 4 | 4 | 7061.023 | 274128 |

| Special Requirements | |
|---|---|
| | Target Of Opportunity Response Time 14 Days |

## Observation

**Proposal 11933, Observation 6: NIRCam 30d**

Fri Mar 13 22:05:25 GMT 2026

**Diagnostic Status: Warning**

Observing Template: NIRCam Imaging

## Diagnostics

(NIRCam 30d (Obs 6)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Generic Targets

| # | Name | Criteria | Description |
|---|---|---|---|
| (1) | Kilonova | The kilonova transient we will target | |

## Template

| Module | Subarray | Target Placement |
|---|---|---|
| ALL | FULL | Module gap (large extended source) |

## Dithers

| # | Primary Dither Type | Primary Dithers | Subpixel Dither Type | Dither Size | Subpixel Positions |
|---|---|---|---|---|---|
| 1 | INTRAMODULEX | 4 | STANDARD | | 1 |

## Spectral Elements

| # | Short Filter | Long Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Dithers | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|
| 1 | F115W | F277W | SHALLOW4 | 10 | 1 | 4 | 4 | 2104.407 | 274128 |
| 2 | F150W | F356W | SHALLOW4 | 10 | 1 | 4 | 4 | 2104.407 | 274128 |
| 3 | F200W | F444W | SHALLOW4 | 10 | 1 | 4 | 4 | 2104.407 | 274128 |

## Special Requirements

Target Of Opportunity Response Time 14 Days

## Observation

**Proposal 11933, Observation 7: MIRI LRS 30d**

Fri Mar 13 22:05:25 GMT 2026

**Diagnostic Status: Warning**

Observing Template: MIRI Low Resolution Spectroscopy

## Diagnostics

(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Generic Targets

| # | Name | Criteria | Description |
|---|---|---|---|
| (1) | Kilonova | The kilonova transient we will target | |

## Acquisition

| # | Target | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 Kilonova | F560W | FASTGRPAVG16 | 10 | 1 | 1 | 444.006 | 274128 |

## Template

| Subarray | Obtain Verification Image? |
|---|---|
| FULL | false |

## Dithers

| # | Dither Type | No. Spectral Steps | Spectral Step Offset | No. Spatial Steps | Spatial Step Offset |
|---|---|---|---|---|---|
| 1 | ALONG SLIT NOD | | | | |

## Spectral Elements

| # | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Exposures/Dith | Total Dithers | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|
| 1 | FASTR1 | 100 | 30 | 60 | 1 | 2 | 16811.192 | 274128 |

| Special Requirements | Target Of Opportunity Response Time 14 Days |
|---|---|

## Observation

**Proposal 11933, Observation 8: MIRI Imaging 30d**

Fri Mar 13 22:05:25 GMT 2026

**Diagnostic Status: Warning**

Observing Template: MIRI Imaging

## Diagnostics

(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Generic Targets

| # | Name | Criteria | Description |
|---|---|---|---|
| (1) | Kilonova | The kilonova transient we will target | |

## Template

| Subarray |
|---|
| FULL |

## Dithers

| # | Dither Type | Starting Point | Number of Points | Points | Starting Set | Number of Sets | Optimized For | Direction | Pattern Size |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2-Point | | | | | | | | DEFAULT |

## Spectral Elements

| # | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Exposures/Dith | Dither | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | F560W | FASTR1 | 80 | 4 | 1 | Dither 1 | 2 | 8 | 1792.676 | 274128 |
| 2 | F770W | FASTR1 | 80 | 4 | 1 | Dither 1 | 2 | 8 | 1792.676 | 274128 |
| 3 | F1000W | FASTR1 | 80 | 4 | 1 | Dither 1 | 2 | 8 | 1792.676 | 274128 |
| 4 | F1280W | FASTR1 | 80 | 4 | 1 | Dither 1 | 2 | 8 | 1792.676 | 274128 |

## Special Requirements

Target Of Opportunity Response Time 14 Days

## Observation

**Proposal 11933, Observation 9: NIRSpec 60d**

Fri Mar 13 22:05:25 GMT 2026

**Diagnostic Status: Warning**

Observing Template: NIRSpec Fixed Slit Spectroscopy

## Diagnostics

(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Generic Targets

| # | Name | Criteria | Description |
|---|---|---|---|
| (1) | Kilonova | The kilonova transient we will target | |

## Acquisition

| # | Target | TA Method | Subarray | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | SAME | WATA | FULL | CLEAR | NRSRAPIDD6 | 3 | 1 | 1 | 171.788 | 274128 |

## Template

| HFF Readout Mode | Slit | Subarray |
|---|---|---|
| false | S400A1 | FULL |

## Dithers

| # | Primary Dither Positions | Sub-Pixel Pattern |
|---|---|---|
| 1 | 2 | SPATIAL |

## Spectral Elements

| # | Grating/Filter | Slit | Readout Pattern | Groups/Int | Integrations/Exp | # | Autocal | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | PRISM/CLEAR | S400A1 | NRSIRS2 | 12 | 6 | 1 | NONE | 4 | 24 | 21358.135 | 274128 |

| Special Requirements | |
|---|---|
| | Target Of Opportunity Response Time 14 Days |

## Observation

**Proposal 11933, Observation 10: NIRCam 60d**

Fri Mar 13 22:05:25 GMT 2026

**Diagnostic Status: Warning**

Observing Template: NIRCam Imaging

## Diagnostics

(NIRCam 60d (Obs 10)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.

(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Generic Targets

| # | Name | Criteria | Description |
|---|---|---|---|
| (1) | Kilonova | The kilonova transient we will target | |

## Template

| Module | Subarray | Target Placement |
|---|---|---|
| ALL | FULL | Module gap (large extended source) |

## Dithers

| # | Primary Dither Type | Primary Dithers | Subpixel Dither Type | Dither Size | Subpixel Positions |
|---|---|---|---|---|---|
| 1 | INTRAMODULEX | 4 | STANDARD | | 1 |

## Spectral Elements

| # | Short Filter | Long Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Dithers | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|
| 1 | F115W | F277W | MEDIUM8 | 8 | 1 | 4 | 4 | 3349.872 | 274128 |
| 2 | F150W | F356W | MEDIUM8 | 8 | 1 | 4 | 4 | 3349.872 | 274128 |
| 3 | F200W | F444W | MEDIUM8 | 8 | 1 | 4 | 4 | 3349.872 | 274128 |

## Special Requirements

Target Of Opportunity Response Time 14 Days

## Observation

**Proposal 11933, Observation 11: MIRI Imaging 60d**

Fri Mar 13 22:05:25 GMT 2026

**Diagnostic Status: Warning**

Observing Template: MIRI Imaging

## Diagnostics

(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Generic Targets

| # | Name | Criteria | Description |
|---|---|---|---|
| (1) | Kilonova | The kilonova transient we will target | |

## Template

| Subarray |
|---|
| FULL |

## Dithers

| # | Dither Type | Starting Point | Number of Points | Points | Starting Set | Number of Sets | Optimized For | Direction | Pattern Size |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2-Point | | | | | | | | DEFAULT |

## Spectral Elements

| # | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Exposures/Dith | Dither | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | F560W | FASTR1 | 80 | 12 | 1 | Dither 1 | 2 | 24 | 5389.128 | 274128 |
| 2 | F770W | FASTR1 | 80 | 12 | 1 | Dither 1 | 2 | 24 | 5389.128 | 274128 |
| 3 | F1000W | FASTR1 | 80 | 12 | 1 | Dither 1 | 2 | 24 | 5389.128 | 274128 |
| 4 | F1280W | FASTR1 | 80 | 12 | 1 | Dither 1 | 2 | 24 | 5389.128 | 274128 |

## Special Requirements

Target Of Opportunity Response Time 14 Days